Fiscal Year:	FY 2024	**Task Last Updated:**	FY 08/04/2023
PI Name:	Mao, Xiao Wen M.D.		
Project Title:	Space Flight Environment Induces Remodeling of Vascular Network and Glia-Vascular Communication in Mouse Retina		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Neurobiology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Zip Code:	92350-0001	**Congressional District:**	31
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2014 Space Biology Flight NNH14ZTT001N
Start Date:	11/01/2014	**End Date:**	10/31/2024
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
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Flight Program:	ISS		
Flight Assignment:	ISS Rodent Research-9 NOTE: End date changed to 10/31/2024 per NSSC information (Ed., 11/9/23) NOTE: End date changed to 10/31/2023 per F. Hernandez/ARC (Ed., 2/27/23) NOTE: End date changed to 10/31/2022 per NSSC (Ed., 12/14/21) NOTE: End date changed to 10/31/2021 per F. Hernandez/ARC (Ed., 11/6/18) NOTE: End date changed to 3/31/2020 per F. Hernandez/ARC (Ed., 4/13/18) NOTE: End date changed to 4/30/2019 per F. Hernandez/ARC (Ed., 6/27/17)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Gridley, Daila Ph.D. (Loma Linda University) Pecaut, Michael Ph.D. (Loma Linda University) Sun, Shu-Wei Ph.D. (Loma Linda University)		

Grant/Contract No.:	NNX15AB41G
Performance Goal No.:	
Performance Goal Text:	
Task Description:	The eye is a unique organ because it is relatively unprotected and constantly exposed to atmospheric oxygen, background radiation, environmental chemicals, and physical abrasion. A recent report shows that more than 30% of the astronauts returning from space were diagnosed with eye problems that can cause headaches and blurry vision. Our Preliminary studies from mice that had been subjected to space flights showed that environmental conditions during space travel lead to oxidative stress and induce adverse microvessel remodeling in the retina. However, our knowledge about the pathophysiological process from adaptive response to irreversible oxidative damage in the retina during and after space flight exposure and the underlying cellular mechanism(s) of these disturbances is very limited and inconsistent. The objectives of this application are to characterize the impact of space flight on female mouse retinal vasculature and tissue remodeling and to study the role of mitochondria in regulating oxidative stress-induced cellular damage and cell-cell interactions in the function of blood-retina-barrier (BRB). Our study will also determine the extent to which an antioxidant metalloporphyrin protects against space flight environment-induced changes in retinal neurovascular coupling and retinal function. The hypothesis will be tested in the specific aims with the space flight experiments on the International Space Station (ISS) and compared to that of ground-based groups using matured female mice. For flight studies, there will be two experiments over two flights, one for in-flight adaptation study on the ISS and one for studying re-adaptation after live animal return to Earth following long-term ISS mission. Selected groups of mice will be treated intraperitoneally with an antioxidant a week before launch and weekly during space flight. Animals will be sacrificed in orbit at 14 and 30 days after launch or 30 days and up to 6 months after return alive on Earth. Together, our unique, integrative, and quantitative activities with advanced MRI (magnetic resonance imaging) imaging techniques and retinal functional testing using visual electroretinography will provide insight into the molecular mechanisms and pathways of space flight condition-induced mitochondrial oxidative damage on retinal BRB integrity, vascular remodeling, and retinal function. The data will provide unique criteria for risks of functional detriments associated with space flight. Our study will also lead to new information to understand the causes and possible treatments of similar Earth-based neurovascular-related diseases and retinal disorders and will point toward potential countermeasures.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our proposed study from space flight mission to ISS will provide a unique opportunity to evaluate the effects of space hazards and mechanisms of response and remodeling of the ocular system. Our study will also lead to new information to understand the causes and possible treatments of similar Earth-based neurovascular-related diseases and retinal disorders and will point toward potential countermeasures.
Task Progress:	Ten-week-old adult C57BL/6 male mice were flown aboard the International Space Station (ISS) on SpaceX-24 over 35 days and returned to Earth alive. The mice received weekly injection of a superoxide dismutase mimic, MnTnBuOE-2-PyP 5+ (BuOE) before launch and during their stay aboard the ISS. Some of the mice from this return group were sent to Loma Linda University (LLU) to re-adapt to Earth's gravity for 4 months. Mice from on-orbit dissection have been returned to LLU for tissue process. Tissue collection and data analysis is still actively ongoing. Fixed ocular and brain samples will be used for histochemical analysis for identifying biomarkers for oxidative stress, apoptosis, and blood-retina-barrier (BRB) function. Frozen retina and brain samples have been used for genomics, proteomics, and metabolomics analysis. The intraocular pressure (IOP) measurements were similar between pre-and post-flight groups with no significant differences between BuOE-treated and saline controls. Immunofluorescence evaluation showed increases in retinal oxidative stress and apoptotic cell death after spaceflight. BuOE treatment significantly decreased the level of the oxidative stress biomarker. Electroretinogram (ERG) data showed that the average amplitudes of the a- and b-wave were significantly decreased 39% and 32% by spaceflight respectively, compared to that of habitat ground controls. The genomics study also showed from subsequent murine retina tissues many genes, pathways, and epigenetic modifications consistent with therapeutic potential of BuOE. Spaceflown murine BuOE treated samples show differentially expressed genes relative to saline controls that reached significance (adjusted p-value < 0.05) and included genes Gpx3 and Crhbp, which are related to protection against cell oxidative damage and cellular response to organonitrogen compounds. [Ed. Note: Per the Principal Investigator, four brain regions were selected for Digital Spatial Profiling: Cortex, Frontal Cortex, Corunu Ammonis 1 (CA), and Dentate Gyrus (DG).] Digital Spatial Profiling (DSP) of mouse brains subjected to either spaceflight, or grounded controls, also revealed hundreds of differentially expressed genes (DEGs) due to spaceflight in all four regions, with CA having the highest number of DEGs. Each brain region showed a distinct transcriptomic response, while some DEGs were common to two or more regions. A pathway analysis revealed distinct pathways sensitive to oxidative stress, as well as metabolic pathways, to be enriched in each of the brain regions. The data indicate, for the first time, the spaceflight condition may induce bipolar cell damage and dysfunction.
Bibliography Type:	Description: (Last Updated: 07/10/2026)
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